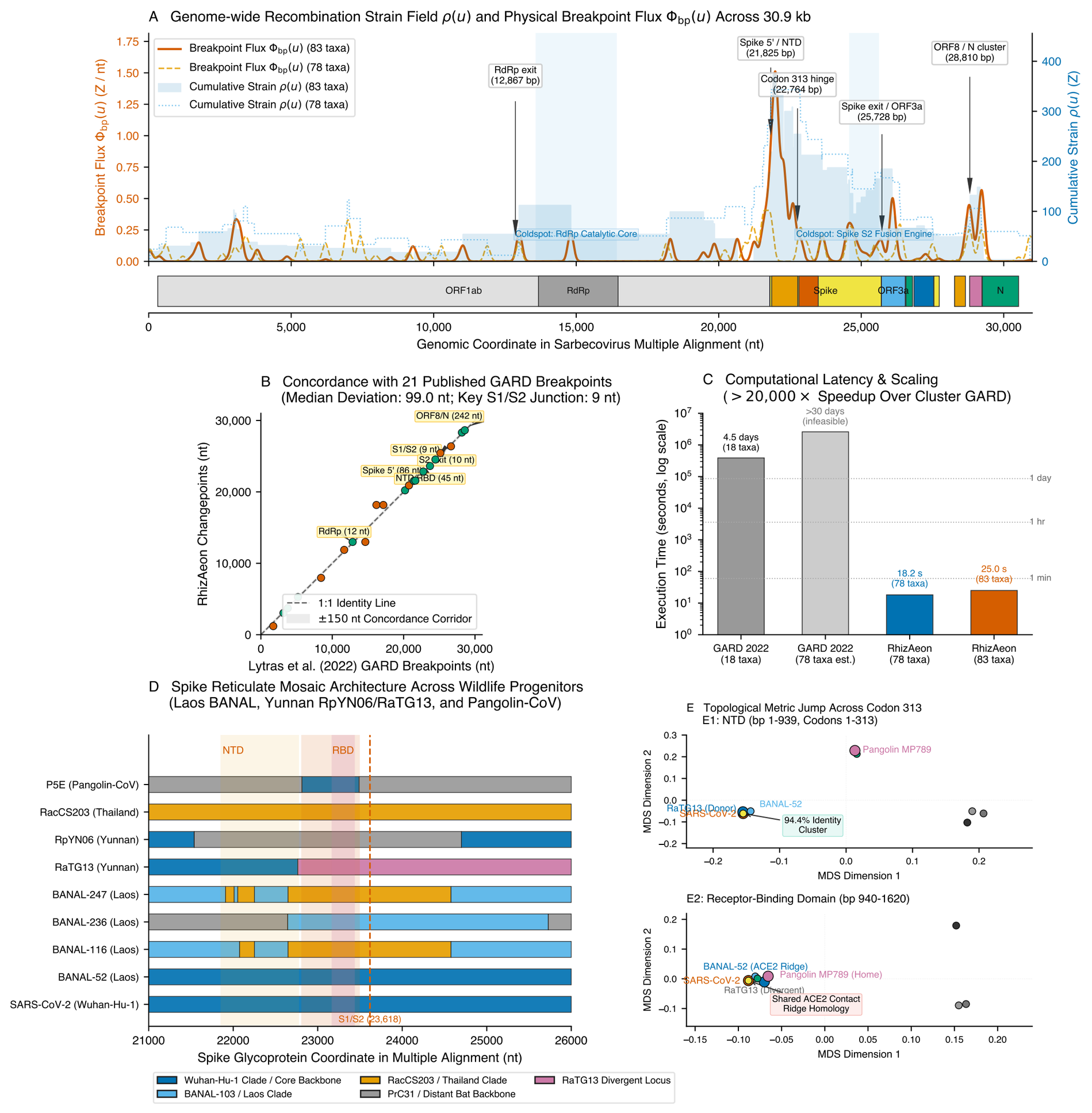

A Genome-wide Recombination Strain Field ρ(u) and Physical Breakpoint Flux Φbp(u) Across 30.9 kb
Breakpoint Flux Φbp(u) (Z / nt)
Cumulative Strain ρ(u) (Z)
Breakpoint Flux Φbp(u) (83 taxa)
Breakpoint Flux Φbp(u) (78 taxa)
Cumulative Strain ρ(u) (83 taxa)
Cumulative Strain ρ(u) (78 taxa)
RdRp exit (12,867 bp)
Spike 5′ / NTD (21,825 bp)
Codon 313 hinge (22,764 bp)
Spike exit / ORF3a (25,728 bp)
ORF8 / N cluster (28,810 bp)
Coldspot: RdRp Catalytic Core
Coldspot: Spike S2 Fusion Engine
ORF1ab
RdRp
Spike
ORF3a
N
Genomic Coordinate in Sarbecovirus Multiple Alignment (nt)
B Concordance with 21 Published GARD Breakpoints
(Median Deviation: 99.0 nt; Key S1/S2 Junction: 9 nt)
RhizAeon Changepoints (nt)
Lytras et al. (2022) GARD Breakpoints (nt)
ORF8/N (242 nt)
S1/S2 (9 nt)
S2 exit (10 nt)
Spike 5′ (86 nt)
NTD-RBD (45 nt)
RdRp (12 nt)
1:1 Identity Line
±150 nt Concordance Corridor
C Computational Latency & Scaling
(> 20,000 × Speedup Over Cluster GARD)
Execution Time (seconds, log scale)
4.5 days
(18 taxa)
>30 days
(infeasible)
18.2 s
(78 taxa)
25.0 s
(83 taxa)
1 day
1 hr
1 min
GARD 2022
(18 taxa)
GARD 2022
(78 taxa est.)
RhizAeon
(78 taxa)
RhizAeon
(83 taxa)
D Spike Reticulate Mosaic Architecture Across Wildlife Progenitors
(Laos BANAL, Yunnan RpYN06/RaTG13, and Pangolin-CoV)
NTD
RBD
P5E (Pangolin-CoV)
RacCS203 (Thailand)
RpYN06 (Yunnan)
RaTG13 (Yunnan)
BANAL-247 (Laos)
BANAL-236 (Laos)
BANAL-116 (Laos)
BANAL-52 (Laos)
SARS-CoV-2 (Wuhan-Hu-1)
S1/S2 (23,618)
Spike Glycoprotein Coordinate in Multiple Alignment (nt)
Wuhan-Hu-1 Clade / Core Backbone
BANAL-103 / Laos Clade
RacCS203 / Thailand Clade
PrC31 / Distant Bat Backbone
RaTG13 Divergent Locus
E Topological Metric Jump Across Codon 313
E1: NTD (bp 1-939, Codons 1-313)
MDS Dimension 2
MDS Dimension 1
Pangolin MP789
BANAL-52
RaTG13 (Donor)
SARS-CoV-2
94.4% Identity
Cluster
E2: Receptor-Binding Domain (bp 940-1620)
BANAL-52 (ACE2 Ridge)
Pangolin MP789 (Home)
SARS-CoV-2
RaTG13 (Divergent)
Shared ACE2 Contact
Ridge Homology